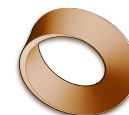
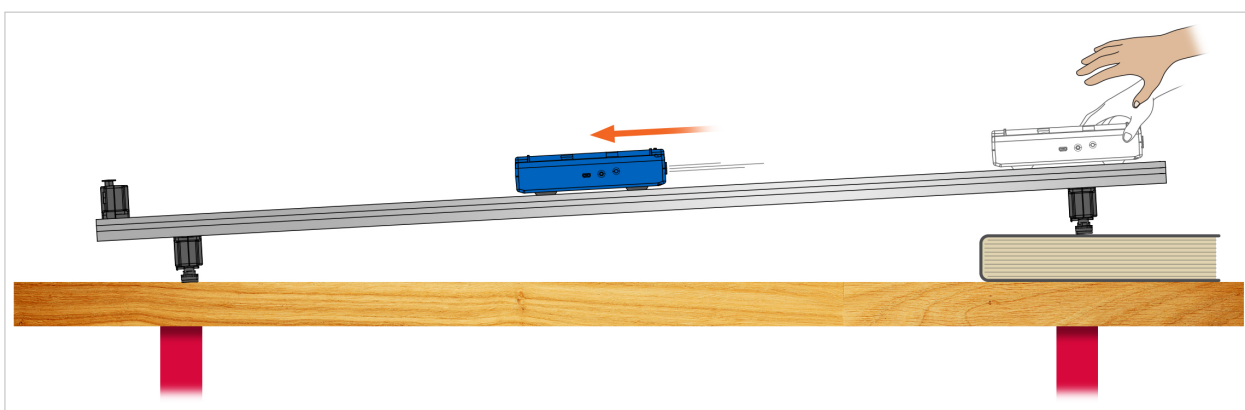


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Class: _____
Date: _____



Mr. Rawson • GCSE Physics

Kinematics 3: Acceleration Down a Ramp — **ANSWERS**



Release the cart from the top and let it run down. Equipment illustration courtesy of PASCO scientific.

Set up

1. Keep the same cart and the same laptop graphs as Experiment 1 — position and velocity, both against time.
2. Put **one book** under one end of the track. A small slope is enough. A steep one just means the cart is gone before the laptop has drawn anything.
3. Fit the **end stop at the bottom** of the track. The cart must have something to hit that is not the floor.
4. Turn the cart so the **+X arrow points down the slope**. Rolling down is now the positive direction.
5. Practice the release once, with no recording: hold the cart at the top, **let go without pushing**, and let it run into the end stop.

Let go — do not push. The whole experiment is about what gravity does to the cart on its own. A push at the start puts a jump in the graph that is nothing to do with the ramp, and you will be able to see it.

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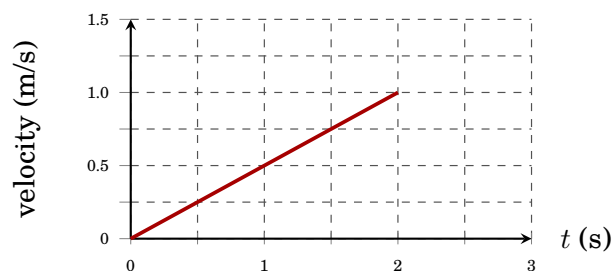
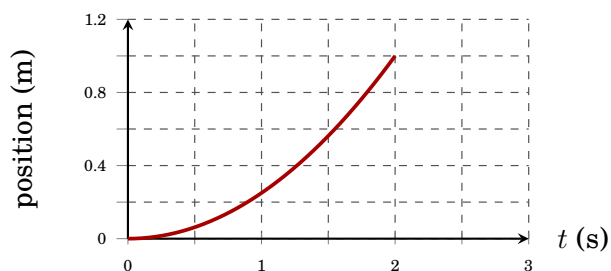
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The runs

Run each one, then sketch what the screen showed. Sketch what you actually see, not what you think it ought to look like.

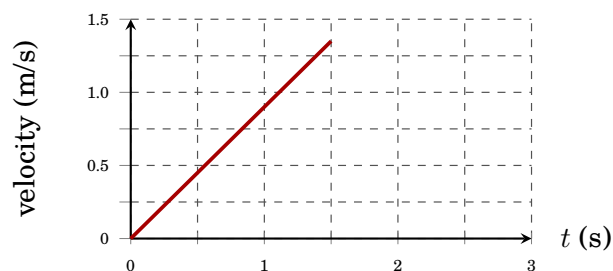
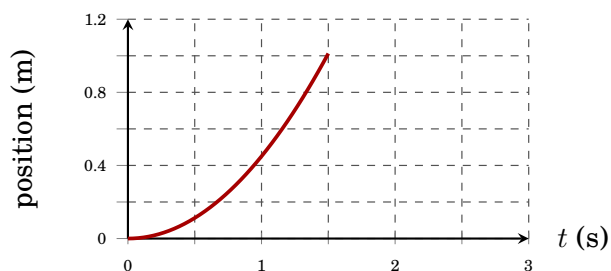
Run 1. One book. Hold at the top, start recording, let go.

Sketch the position–time graph and the velocity–time graph.



Run 2. Two books — a steeper ramp. Same release.

Sketch the position–time graph and the velocity–time graph.



Your results

Key concept

Use SPARKvue's  **Linear Fit** on the **velocity–time** line. It gives you

$$y = mx + b$$

where m is the **slope** and b is the **y-intercept** — the velocity the cart had at the very start of the run.

In Experiment 1 every velocity–time slope came out as zero. **Watch what m does here.**

Fit only the part where the cart is running freely — leave out your hand at the start and the end stop at the finish.

Run	Slope m of v - t	y -intercept b of v - t / m s^{-1}
1 one book	<i>a positive number</i>	<i>≈ 0 — released from rest</i>
2 two books	<i>a bigger positive number</i>	<i>≈ 0 — released from rest</i>
3 up and back	<i>a negative number</i>	<i>positive — the speed of the push</i>

Now compare the two velocity lines from runs 1 and 2.

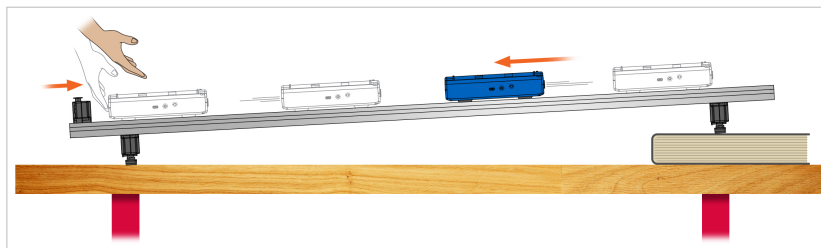
1) **Describe** what *both* velocity lines have in common, and how they differ.

Both are straight lines that start at zero and tilt upwards — the cart speeds up steadily in both. The two-book line tilts more steeply, because on the steeper ramp the cart gains speed faster.

2) **Explain** what the position–time line was doing in run 1, and why it was not straight.

It curved upwards, getting steeper as time went on. The cart covered more distance in each successive second because it was going faster and faster, and a curve that keeps getting steeper is what a speeding-up object looks like.

Run 3 — up the ramp, and back down



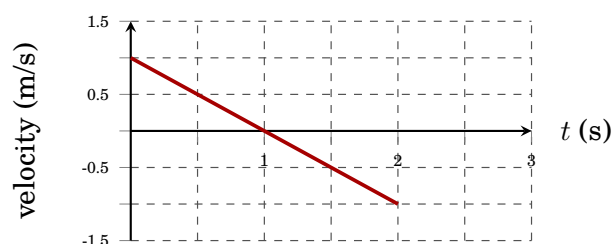
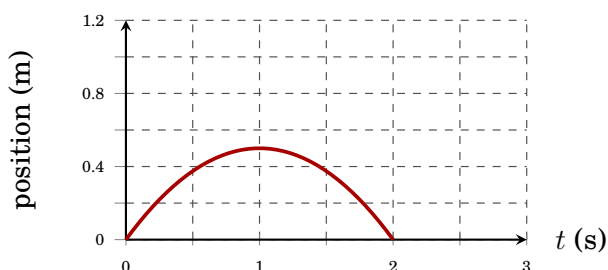
Push it up the slope and let it come back. Illustration: PASCO scientific.

Put the ramp back to **one book**. Start recording, then push the cart **up** the slope from the bottom and let it roll all the way back down. Stop recording before it reaches the end stop. **Catch it.**

For this run only, turn the cart so +X points up the slope — up is now positive, so your position graph climbs from zero and comes back to it.

Run 3. Pushed up the slope, slows, stops, comes back down.

Sketch the position–time graph and the velocity–time graph.



1) **Identify** the moment the cart was momentarily still. What was the velocity line doing at exactly that moment?

At the highest point of its travel, roughly halfway through the recording. The velocity line crossed the zero line there — it passed through zero on its way from positive to negative. On the position graph this is the top of the curve.

2) **Explain** why the velocity line crosses the zero line in this run, when it never did in Experiment 1.

The cart changed direction. It was travelling up the ramp (positive) and ended up travelling down it (negative), and the only way a velocity line can get from above the zero line to below it is by passing through zero.

3) **Describe** the shape of the velocity line over the whole run, in one sentence.

One straight line, tilting downwards the whole way through — it does not bend or kink at the moment the cart turns around.

*Students' own traces will differ — the **shape** is what is being checked. Real traces show the push at the start of run 3 and a kink where the cart is caught. Expect run 3's velocity line to look slightly less straight than the ideal: friction makes the cart lose speed a little faster going up than it gains coming down.*